

Graph Rewriting and its Confluence

Jean-Pierre Jouannaud

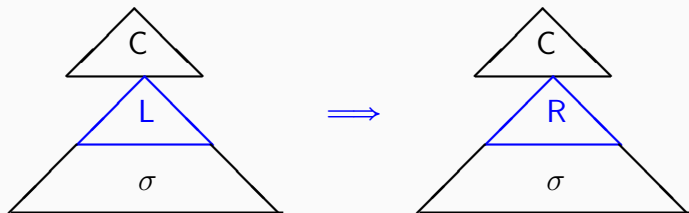
Joint work with

Nachum Dershowitz and **Fernando Orejas**

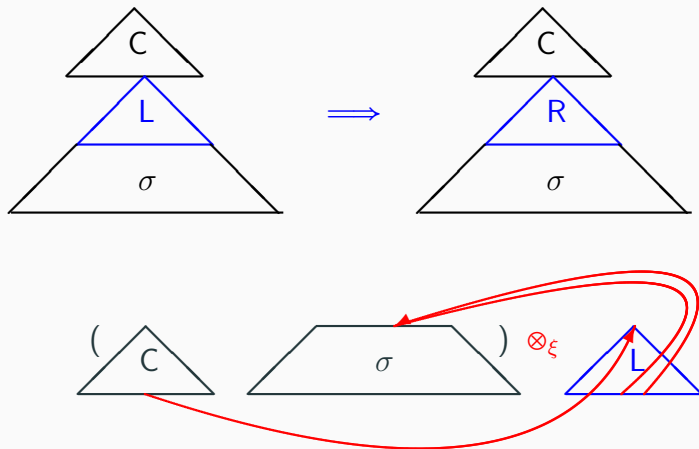
July 17, 2026

Scale term rewriting theory to graphs

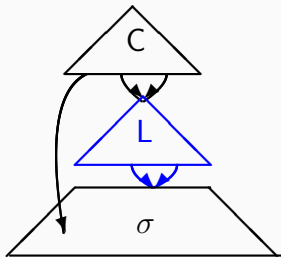
Tree Rewriting and Composition



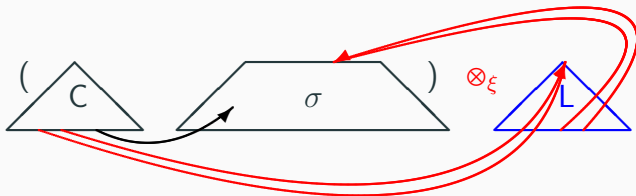
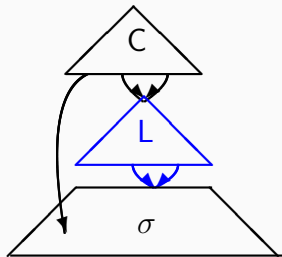
Tree Rewriting and Composition



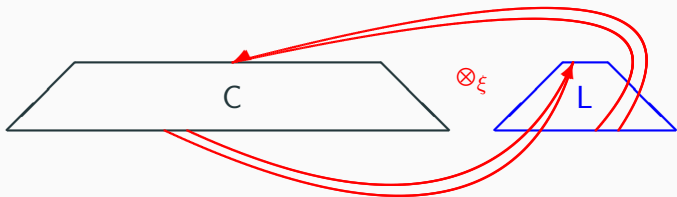
Dag Rewriting and Composition



Dag Rewriting and Composition



Graph Rewriting and Composition



We need multi-rooted multi-graphs with variables labeling some leaves

Graphs

Rooted Graphs

Σ function symbols labeling vertices:   

Ξ variable symbols with arity zero:  

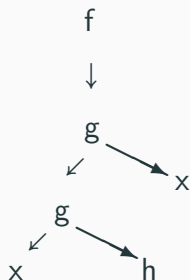
A *compositional graph* consists of

- *vertices* V
- *sprouts* $S \subseteq V$, leaving $I = V \setminus S$ for the *internal vertices*
- *edges* E is a set, an edge is a triple $e : u \rightarrow v$, $u \notin S$, name e , tail u , head v , restrictions E_I and E_S depending on v
- *roots* R is a set; $R(v)$ is the set of roots pointing at $v \in V$
- *labels* $L : S \rightarrow \Xi$
 $L : I \rightarrow \Sigma$

Ground compositional graphs have neither roots nor sprouts

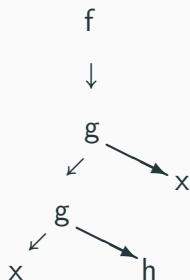
Labels in Σ , possibly indexed, will serve as names for vertices

Example: Tree, Dag, Drag, Graph

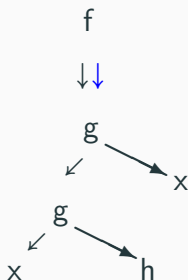


Term

Example: Tree, Dag, Drag, Graph

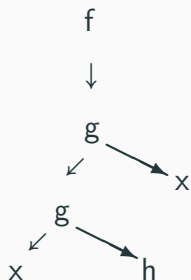


Term

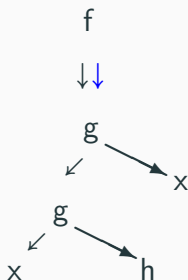


Dag

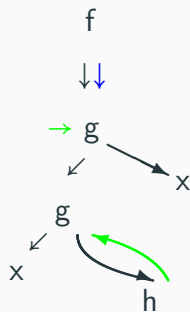
Example: Tree, Dag, Drag, Graph



Term

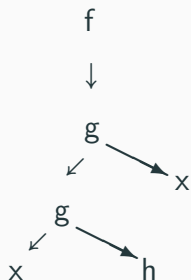


Dag

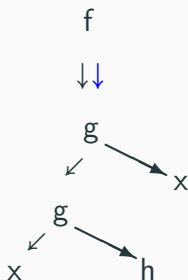


Drag
Term Graph

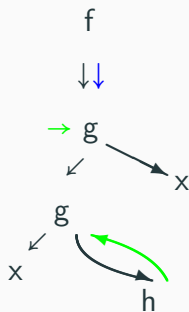
Example: Tree, Dag, Drag, Graph



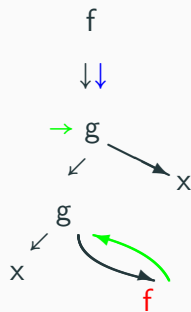
Term



Dag



Drag
Term Graph



Compositional
Graph

Morphisms on compositional graphs

Miscellaneous

- *Subgraphs*: generated by a set of vertices w : $D|_w$
- *Connected component*: set of vertices closed under successors and predecessors, and equal labeling of sprouts
- *Equimorphic graphs*: identical up the name of their vertices
- *Compacted graphs*: no two subgraphs are equimorphic
- *Sharing equivalent graphs*: equimorphic when compacted

Morphisms

Morphism $o : D = \langle V, S, E, R, L \rangle \hookrightarrow D' = \langle V', S', E', R', L' \rangle$

- $o_V : V \rightarrow V'$, the *vertex map*
- $o_E : E \rightarrow E'$, the *edge map*
- $o_R : R \rightarrow R'$, the *root to root map*
- $o_{R,E} : R \rightarrow E'$, the *root to edges map*
 1. o_V restricts to mappings from I to I' , and preserves labels;
 2. o_V restricts to mappings from isolated sprouts to isolated sprouts;
 3. $o_E, o_R, o_{R \rightarrow E}$ commute with o_V
 4. o preserves sharing equivalent subgraphs

Notation: o_{E_I}, o_{E_S} restrictions of o_E to internal- and demi-edges

Matching morphisms

Matching morphism $o : D \hookrightarrow D'$ is a morphism such that

- o_V restricts injectively to I and isolated sprouts
- o_E , $o_{R,R}$ and $o_{R,E}$ are injective
- $\text{Im}(o_{E_S}) \subseteq \text{Im}(o_{R,E})$

We say that D is *embedded* into D' and write $o : D \hookrightarrow D'$.

An *injection* is a matching morphism whose vertex map is the identity on internal vertices and isolated sprouts. An *isomorphism* is a matching morphism whose vertex, edge, and root maps are all bijective ($o_{R,E}$ being then empty).

Operations on compositional graphs

Parallel and cyclic compositions

Two graphs D, D' are *compatible* if

- **shared** internal vertices **share** their label
- if some variable x is **shared**, some **shared** sprout is labeled x
- **shared** edges **share** their sources and their targets
- **shared** roots **share** their vertex

Parallel and cyclic compositions

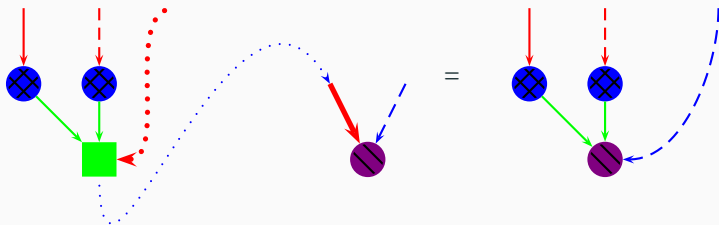
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If D, D' are compatible,

- their *parallel composition* is $D \oplus D'$
- their *cyclic composition* wrt switchboard ξ is $D \otimes_{\xi} D'$
- cyclic composition is obtained by *wiring* $D \oplus D'$ wrt ξ , where ξ tells you where to branch sprouts from one side to roots of the other side

Wiring



A **wire** $s \rightsquigarrow u$ is a pair made of a sprout s and an vertex $u \neq s$

A **switchboard** ξ for D, D' is a set of wires $(\xi_{D \rightarrow D'}, \xi_{D' \rightarrow D})$ s.t. equally labeled sprouts are replaced by vertices yielding sharing equivalent subgraphs in the wired graph

Wiring eats as many roots of the wire's target as the number of predecessors of the wire's origin, whose roots are lost.

Major properties

Algebra of rooted graphs: parallel and cyclic compositions are associative, commutative, and have the empty graph as neutral element.

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Matching Theorem: Assume $o : L \hookrightarrow G$ is a matching morphism such that L is a *pattern*, that is, a *graph without isolated sprout*. Then, there exists a *rewriting extension* (C, ξ)

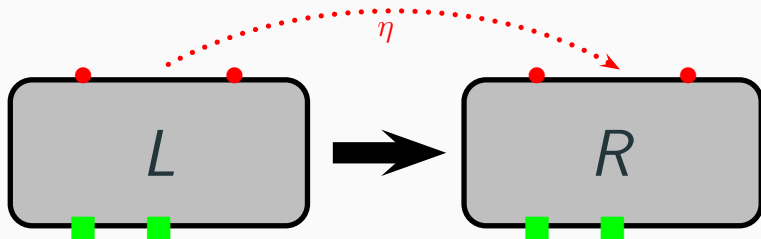
- C is a linear compositional graph
- ξ_C is surjective on the roots of L
- ξ_L is total

such that

$$G = C \otimes_{\xi} L$$

Rewriting

Rules as (pairs of) drags



- L, R are drags, the *pattern* L contains no isolated sprout
- all variables have an occurrence in the pattern
- *root-map* $\eta : \mathcal{V}_L \rightarrow \mathcal{V}_R$ that *preserves roots in number*
- rewriting: $C \otimes_{\xi} L \longrightarrow C \otimes_{\xi'} R$
- $(C, \xi), (C, \xi')$ are *rewriting extension* of L and R s.t.
 C, C' are sharing-equivalent and share no vertex with L
- root-map η ensures the absence of dangling edges

Drugs: Current research status

Results

- Term rewriting is a particular case of drag rewriting
- Termination: a recursive path ordering for drags
(can be total on ground drags)
- Unification: a square complexity propagation algorithm for drags by marking pairs of corresponding vertices
- Local confluence for drags by joinability of critical pairs

Results

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Proof sketch: let U rewrite with rules $L \rightarrow R$ and $G \rightarrow D$.

- overlapping case uses drag unification for computing critical pairs, and lifting their joinability by monotonicity
- non-overlapping case thanks to AC-property of $\otimes$:

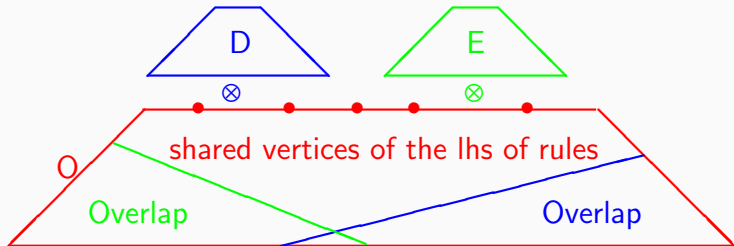
$$U = C \otimes (L \oplus G) = C \otimes (L \otimes G)$$

$$U = (C \otimes L) \otimes G \longrightarrow (C \otimes L) \otimes D = (C \otimes D) \otimes L \longrightarrow (C \otimes D) \otimes R = V$$

$$U = (C \otimes G) \otimes L \longrightarrow (C \otimes G) \otimes R = (C \otimes R) \otimes G \longrightarrow (C \otimes R) \otimes D = V$$

Overlapping case

C



Minimal overlapping drag is $(D \oplus E) \otimes O$

Drag $U = C \otimes ((D \oplus E) \otimes O)$

Unification

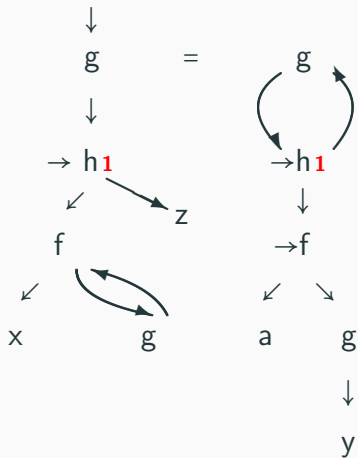
Drag unification

Given drags U, V , we call **partner vertices** two lists L_U, L_V of equal length of internal vertices of U and V , such that no two vertices $u, u' \in L_U$ (resp., $v, v' \in L_V$) are related by a path.

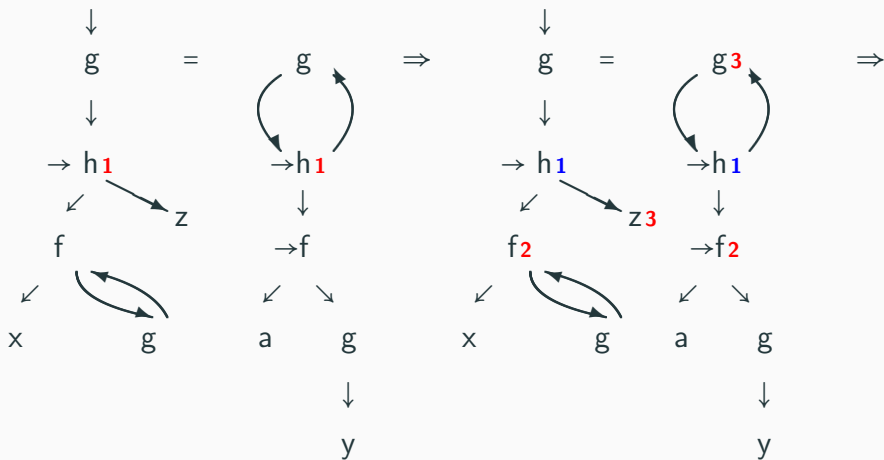
A drag **unification problem**, $U[\bar{u}] = V[\bar{v}]$, is a pair (U, V) of connected patterns that have been renamed apart, together with partner vertices $P = (\bar{u}, \bar{v})$. A **solution** (or *unifier*) of the drag unification problem $U[\bar{u}] = V[\bar{v}]$ is a pair of rewriting extensions (C, ξ) and (D, ζ) of U and V respectively, such that $C \otimes_\xi U$ and $D \otimes_\zeta V$ are *identified at* $(\bar{u}, \bar{v})$.

A drag equal modulo renaming to $C \otimes_\xi U$ (hence to $D \otimes_\zeta V$) is called **overlap** of U, V at $(\bar{u}, \bar{v})$.

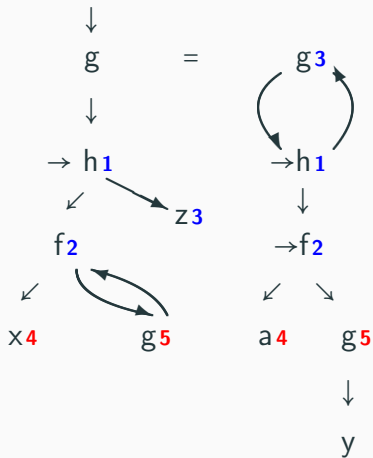
Drag unification rules at work (given L_U, L_V) 1/3



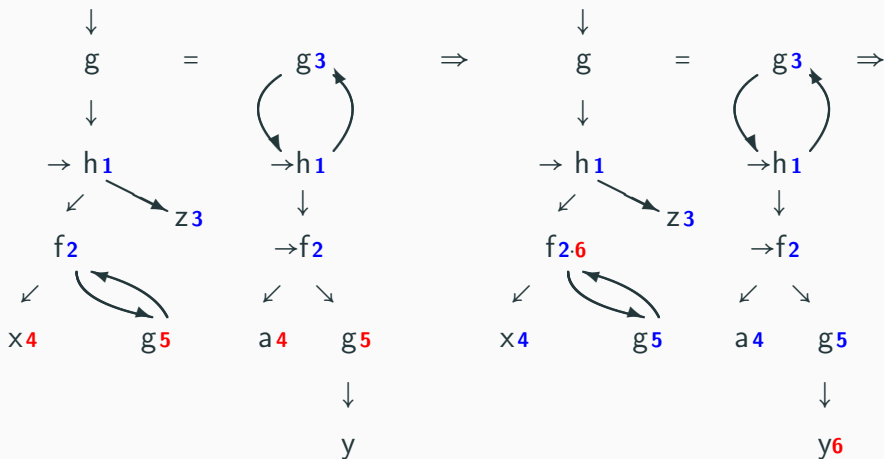
Drag unification rules at work (given L_U, L_V) 1/3



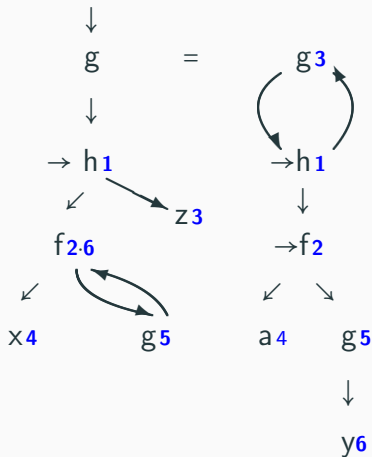
Drag unification rules at work 2/3



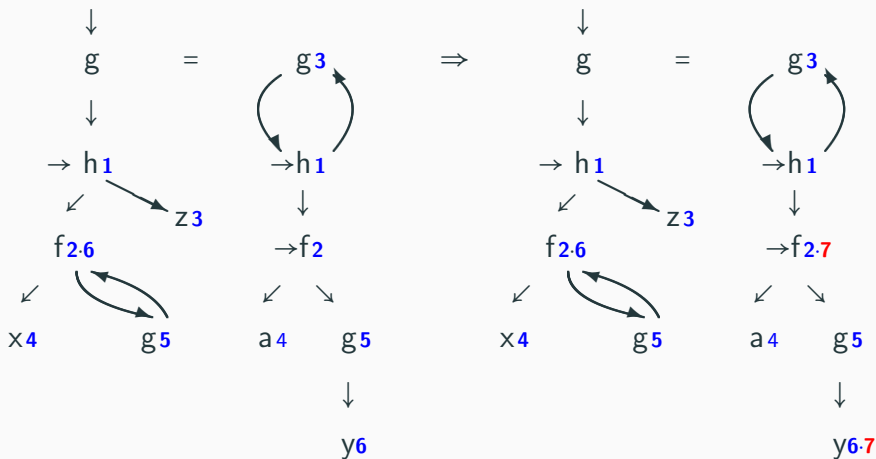
Drag unification rules at work 2/3



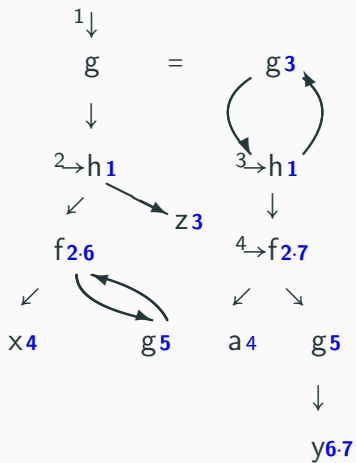
Drag unification rules at work 3/3



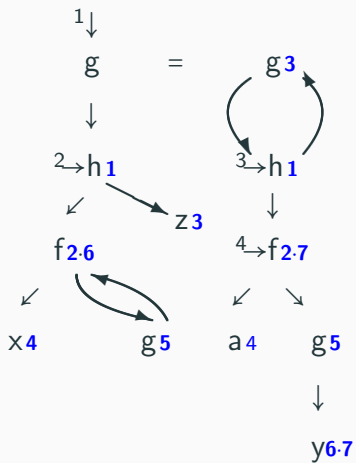
Drag unification rules at work 3/3



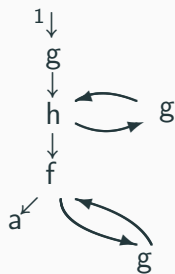
Most general unifier



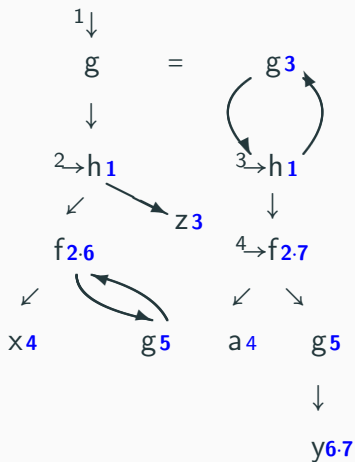
Most general unifier



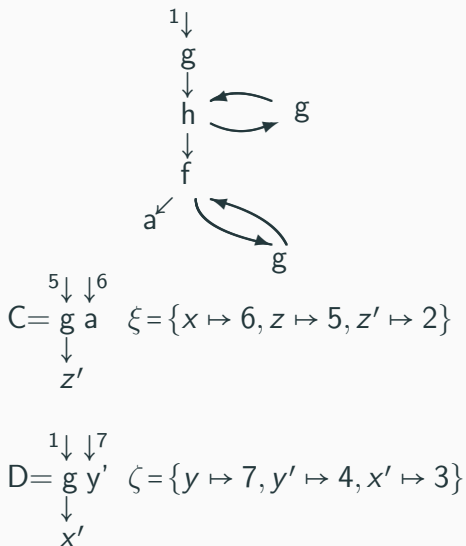
Overlap



Most general unifier



Overlap



Local confluence of compositional graphs

Step 1: reduce (non-deterministically) unification of compositional graphs to drag unification

Step 2: apply local confluence of drags

Step 3: deduce local confluence of compositional graphs

Double Pushout Rewriting

Comparison with DPO

Rule: $L \leftarrow C \hookrightarrow R$ (This is called a *span*)

Match: $L \hookrightarrow G$

Pushout complement: $G \leftarrow C' \leftarrow C$

Pushout: $C' \hookrightarrow G' \leftarrow R$

This construction requires the existence of pushouts and pushout complements.

It works in any adhesive category of graphs with pushouts.

PBR compared with DPO

- *No explicit variables in DPO, explicit variables in PBR*
- *Implicit linear variables via monomorphisms in DPO*
- *DPO: no known way to encode non-linear rewriting*
- *DPO does NOT generalize term rewriting, PBR does*
- *DPO may generate dangling (a source, no target) edges*
- *DPO works uniformly over adhesive categories of graphs*
- *Adhesivity must be checked for each category of graphs*
- *Product defined uniformly for all kinds of rooted graphs*
- *Matching Theorem proved for each category of graphs*
- *The category of ground compositional graphs is adhesive*
- *The category of compositional graphs is not adhesive*
- *DPO: Confluence is undecidable for terminating systems*
- *PBR: Confluence decidable for terminating systems*

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J. of Log. and Alg. Methods in Programming 131 (2023)
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